

Use-Case Flow Specification

PES University – Dept. of CSE | Lab 1: Requirements Engineering & UML Use-Case Modelling

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Use Case Identification

Use Case: UC-03 — Borrow/Reserve Equipment

Primary Actor: Community Member

Supporting Actors: Library Custodian, Depot Locker System

Preconditions

- The Community Member has a valid, authenticated account.
- The requested equipment item exists in the inventory.
- The requested equipment item is currently available.

Main Success Scenario

1. The Community Member searches for the required equipment by category, name, or keyword.
2. The system displays matching equipment items with current availability status.
3. The Community Member selects an available item to borrow.
4. The system verifies the member's borrowing eligibility.
5. The system checks the applicable loan duration limits for the equipment category.
6. The system calculates the required security deposit based on the equipment category.
7. The system displays the deposit amount, loan duration, and expected return date.
8. The Community Member confirms the borrowing request.
9. The system creates a loan record with the member, equipment, deposit, and return date.
10. The system updates the equipment status to "Checked Out" in the inventory.
11. The Depot Locker System updates the physical availability indicator.
12. The system displays a borrowing confirmation to the Community Member.

Alternate Flow

A1 — Member Has Overdue Equipment

At step 4, the system detects that the Community Member has an overdue equipment item.

- 4a. The system prevents the new borrowing request.
- 4b. The system displays the overdue item and borrowing restriction.
- 4c. The system informs the member that borrowing cannot continue until the restriction is resolved.
- 4d. The use case ends without creating a new loan record.

Postconditions

On Main Success Scenario completion:

- A loan record exists associating the member, equipment, deposit, and return date.
- The equipment status is updated to "Checked Out" in the inventory.
- The member's security deposit is recorded as held.
- The physical availability indicator at the Depot Locker is updated.

On Alternate Flow (A1) completion:

- No new loan record is created.
- The equipment item remains available.
- The member is notified of the restriction.